

Motivation

- Leaky Integrate-and-Fire Neurons
 - work by integrating (accumulating) input currents over time to increase a "membrane potential" and "fire" (produce a spike) when this potential crosses a threshold, similar to how a leaky RC circuit charges. After firing, the potential resets. The "leaky" part refers to the fact that the membrane potential also gradually decays or "leaks" back to a resting value over time unless it is being driven by input
- The SCN idea
 - a population of LIF neurons only fire when it reduces the global representation error. This yields sparse, irregular, yet coordinated spikes
 - The paper shows that the same network can implement a linear dynamical system (add a "slow" recurrent term).
 - Kalman filter for optimal state estimation through noisy observations
 - LQR for optimal control to a target $z(t)$
 - The control u is a linear readout of the spikes
 - -> Spiking LQG

Setup

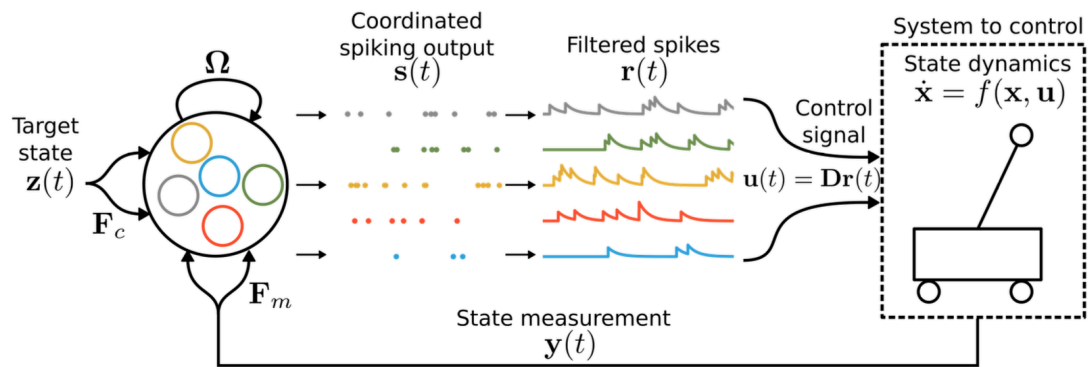


Fig. 1: Controlling dynamical systems with spiking neural networks. A recurrent spiking neural network (left) will emit spikes based on its inputs and connectivity, which are translated into a control signal to control some target system (right). F_c and F_m represent input weights, Ω represent recurrent weights, and D represent read-out weights. By first defining the optimal control solution for the system, and then directly translating the resulting control parameters into these connection weights, we can generate a robust and population-wide coordinated spiking network that accurately controls identified systems.

- x : state
- y : incomplete measurement of the system state
- z : incomplete measurement of the target state
- u : control signal
- s : spiking signal emitted from a recurrent SNN, used to control the state
- r : filtered spikes

